Name:- Darpan N. Rodrane Roll No:- 5004 Div:- 5 Batch:- A1
SAP ID:- 60018250070 branch:- A I & DS

Experiment No 8

Q1 Define transactions and states of transactions

=> A transaction is a logical unit of work made up of one or more SQL statements that read or change the database. It must either complete fully or not happen at all, so the database never ends up half updated. For example, transferring money between two accounts involves a debit and a credit, and both must succeed together. Every transaction follows the ACID properties, which are Atomicity, Consistency, Isolation and Durability.

States of a Transaction

1. Active : The transaction is executing its statements. This is the initial state.

2. Partially committed :- The final statement has executed, but the changes are not yet made permanent.

3. Committed : The changes are saved permanently in the database.

4. Failed : Normal execution cannot continue because of an error or a system problem.

5. Aborted : The transaction is rolled back and the database return to its state before the transaction began. After this the transaction can be restarted or killed.

Sundaram
FOR EDUCATIONAL USE

Q2 write syntax for TCL commands

⇒ 1) START TRANSACTION;:- Begins a transaction. BEGIN also works.

2) COMMIT;:- saves all changes made in the current transaction permanently.

3) ROLLBACK;:- undoes all changes made since the transaction began or since the last commit.

4) SAVEPOINT sp1;:- creates a marker inside a transaction that you can return to later.

5) ROLLBACK TO sp1;:- undoes changes only up to the named savepoint.

6) RELEASE SAVEPOINT sp1;:- Removes a savepoint that is no longer needed.

7) SET TRANSACTION READ WRITE;:- Sets the properties of the transaction, such as readonly or read write.

Sundaram FOR EDUCATIONAL USE